Bengali ASR: five models on FLEURS

920 FLEURS bn_in TEST utterances · one A100-SXM4-40GB · greedy, batch 1, no external LM · ‡ decoded by vLLM

Model	WER	95% CI (clustered)	CER	ms/clip	Params
Conformer Large (hishab)	14.37%	[13.11–15.73]	4.48%	87	121.5M
Voxtral Mini bn-BD ‡	14.73%	[13.60–15.93]	4.78%	438	4,676M
Whisper Medium	16.22%	[15.04–17.45]	5.09%	1,901	763.9M
▸ **ehzawad FastConformer-CTC**	**17.12%**	**[15.95–18.34]**	**5.12%**	**88**	**115.6M**
Wav2Vec2	20.66%	[19.47–21.90]	6.02%	56	315.5M

Test split only. The ehzawad model trained on a corpus that includes the FLEURS train split, and 88 of the 150 validation-split sentences also occur there; the 402 validation recordings are therefore excluded for every model and the table is the 920-utterance test split.

Reading this table

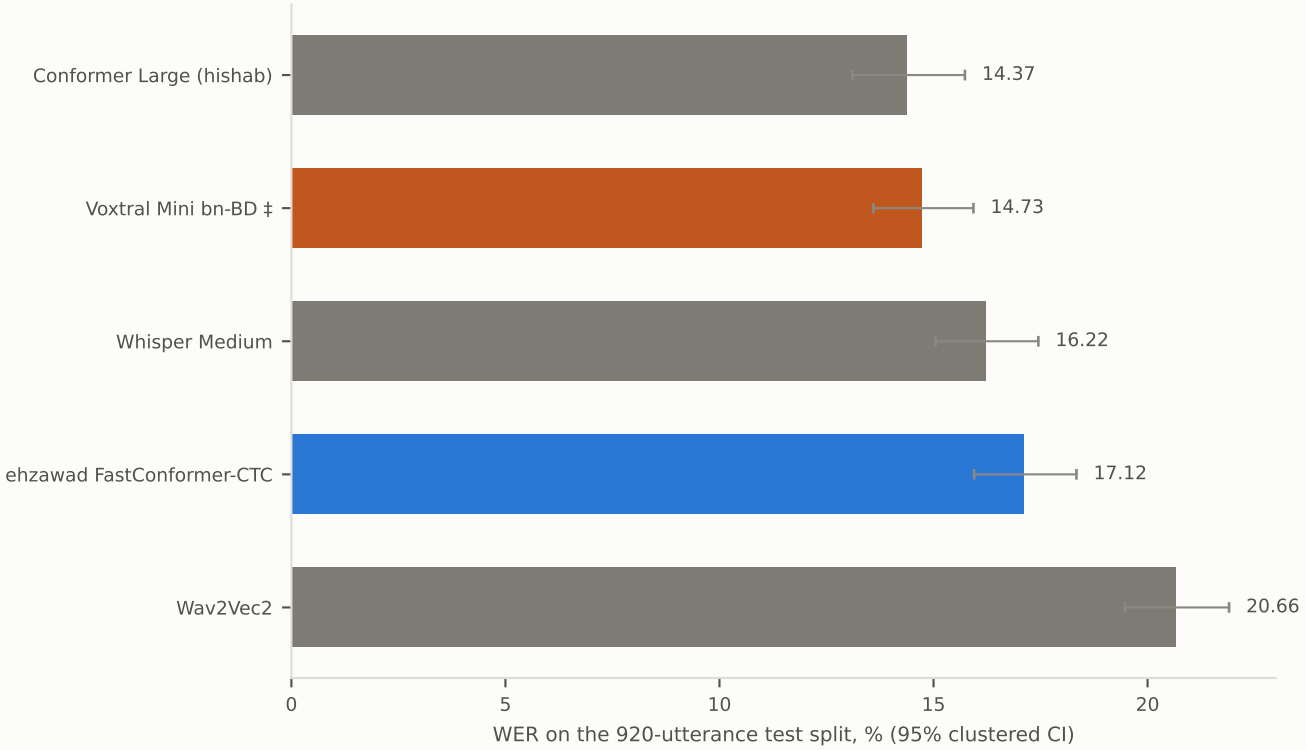
ehzawad/stt_bn_fastconformer_ctc is a FastConformer-CTC large (115.6M parameters, 8x subsampling, 1,024-piece Bengali BPE) trained on 1,692 hours of human-supervised Bengali, CC-BY-SA-4.0. The other three public models are checkpoints from hishab and SayedShaun (mirrors of tugstugi's Whisper and qdv206's wav2vec2 competition models).

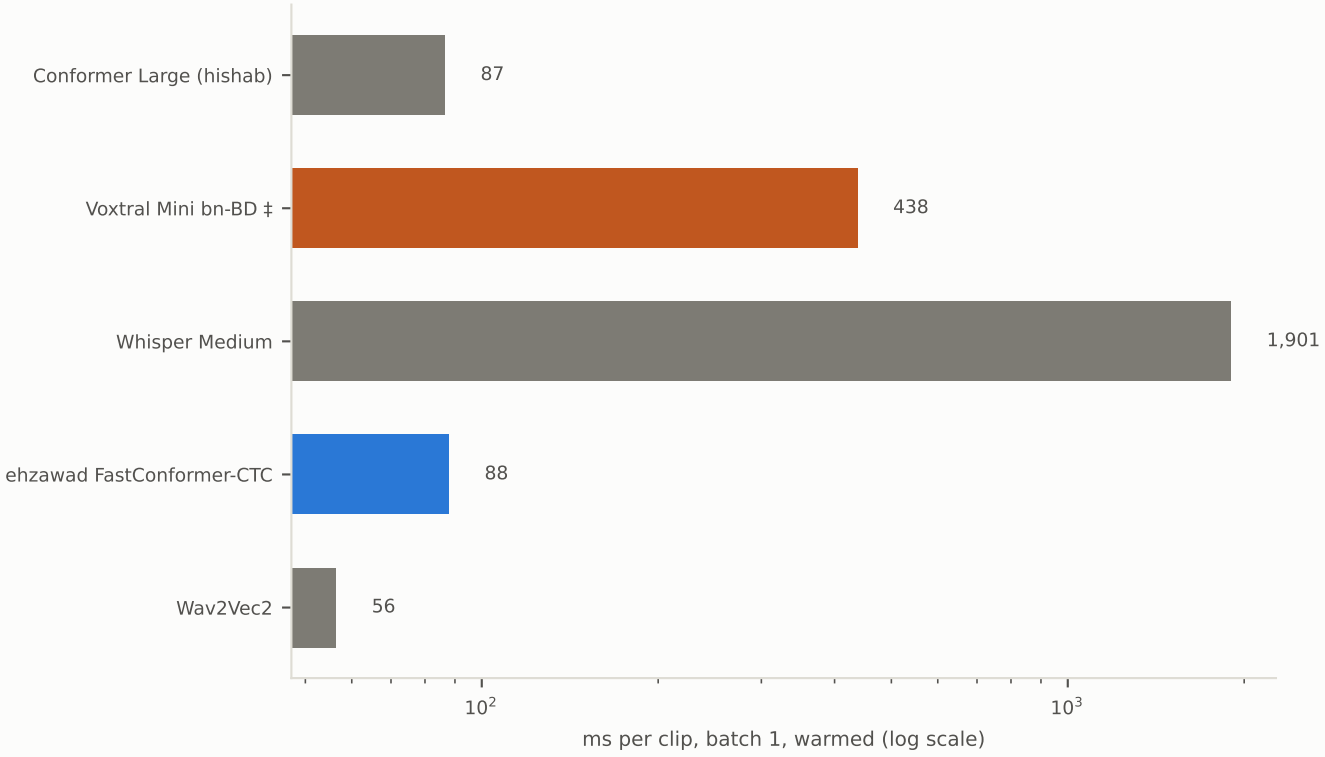
Paired sentence-clustered deltas for the highlighted model (10,000 draws):
vs Whisper Medium +0.90 pp [+0.28, +1.51]
vs Conformer Large +2.74 pp [+2.01, +3.46]

0.9 points behind Whisper Medium at 22x its speed; 2.7 points behind Conformer Large. The highlighted model had also been evaluated on FLEURS test during its own development, so read its FLEURS row as a like-for-like re-measurement and the other corpora as the fresh evidence.

‡ Voxtral Mini bn-BD is a private fine-tune of Voxtral Mini 3B whose card documents no training data. It went through the same harness, asr_core path and scorer, decoded at batch 1 by vLLM 0.31 rather than by PyTorch directly. Against Conformer Large on this split: +0.35 pp [-0.36, +1.06], inconclusive.

FLEURS test: accuracy and speed side by side





The two orderings are not the same ordering. The two NeMo CTC models are within 2% of each other in speed; Whisper Medium costs ~22x their time per clip at batch 1 for its accuracy, and Voxtral Mini ~5x (vLLM with CUDA graphs).

Beyond FLEURS: six more test sets

Frozen seeded 1,000-utterance subsets of the OFFICIAL splits · one model at a time · same backends and scorer

Set	official rows	sampled	audio	ref. words	stratified by
Vaani test	12,695	1,000	1.26 h	10,082	29 state/district strata
SPRING-INX R1	1,879	1,000	2.54 h	19,706	Latin / no-Latin reference
SPRING-INX R2	1,872	1,000	2.61 h	20,353	Latin / no-Latin reference
Kathbath known	2,806	1,000	1.79 h	10,801	speaker gender
Kathbath unknown	1,783	1,000	1.81 h	10,887	speaker gender
IndicVoices conv.	3,603	1,000	1.57 h	12,918	scenario (conversation / extempore)

Model (WER %)	Vaani	SPR R1	SPR R2	Kb known	Kb unk	IV conv	flags
▸ **ehzawad FastConformer-CTC**	**24.67**	**39.23**	**38.42**	**13.38**	**13.43**	**12.61**	**N N N S S S**
Whisper Medium	26.27	39.97	39.70	9.69	9.86	25.00	U U U U U U
Conformer Large (hishab)	33.10	42.46	42.25	9.33	8.79	29.68	N N N S S N
Wav2Vec2	31.28	47.28	47.32	19.07	18.80	33.02	U U U U U U
Voxtral Mini bn-BD ‡	22.08	36.87	40.01	10.77	10.82	13.40	U U U U U U

flags, per set in the column order: N = corpus not in the model's declared training data;
S = trained on a sibling split of the same corpus (Kathbath train; Conformer Large also its validation split); U = training provenance not documented. None of these means "verified clean".
Every cell is a batch-1 decode from this run. IndicVoices conversational is NOT a clean head-to-head for the highlighted model: it trained on the IndicVoices train split (same collection, prompts and transcription conventions, different speakers), which is a large part of its 12.6% there.

What the table says

On the corpora nobody is documented as having trained on, the FLEURS ranking does not hold. Conformer Large, the FLEURS leader at 14.4%, is last on Vaani (33.1%) and fourth on both SPRING sets; its model card lists read and broadcast corpora and neither Vaani nor SPRING-INX. ehzawad/stt_bn_fastconformer_ctc leads the four public models on Vaani and both SPRING sets; Whisper Medium is second on all three at 11-19x the latency of the NeMo CTC models. The private Voxtral Mini fine-tune (‡, U on every set) is lower still on Vaani and SPRING R1 (page 5).

On Kathbath the order flips back: Conformer Large (9.3 / 8.8%) and Whisper (9.7 / 9.9%) lead, Voxtral Mini follows (10.8 / 10.8%) and the highlighted model is fourth (13.4 / 13.4%). Conformer Large trained on Kathbath validation as well as train, Whisper's provenance for Kathbath is unknown (a U cell can hide training overlap), and the highlighted model saw only the train split. Known- vs unknown-speaker Kathbath differ by under a point for every model. On IndicVoices conversational the three other public models read 25-33%; Conformer Large, the one N cell, emitted 7 empty hypotheses there and Wav2Vec2 33. Sentence-clustered 95% intervals for every cell are in the README and summary.

Latency per set, and how the runs were controlled

SPRING note: about half of its reference utterances contain English words in Latin script; no model emitted a single Latin-script word, so those words are errors for every model alike. On the Latin-free half the highlighted model (35.4 / 35.1%) and Whisper Medium (36.0 / 36.6%) are within each other's intervals, and Voxtral Mini reads 32.2 / 37.7% (per-model split in outputs_multiset/summary_multiset.json).

ms per clip, batch 1, warmed (Voxtral Mini through vLLM with CUDA graphs)

Model	Vaani	SPR R1	SPR R2	Kb known	Kb unk	IV conv
▸ **ehzawad FastConformer-CTC**	**81**	**85**	**86**	**81**	**82**	**82**
Whisper Medium	886	1,520	1,579	1,109	1,127	1,028
Conformer Large (hishab)	81	84	85	82	81	82
Wav2Vec2	33	47	46	34	35	38
Voxtral Mini bn-BD ‡	240	409	431	291	290	291

Protocol. Every set is a frozen subset (seed 20260915; proportional stratification; per-file SHA-256 verified before decoding). One model resident at a time on the A100; no other process used the card during any run. Manifests: multiset/; raw predictions and per-run GPU state: outputs_multiset/; harness bench_run_set.py (bench_run.py's backends unchanged; Voxtral Mini through the voxtral_vllm backend); scorer score_multiset.py. Reproduce the scoring from the committed predictions.

Evaluation data: FLEURS (Google, CC-BY-4.0); Vaani transcription part (ARTPARK-IISc, CC-BY-4.0); SPRING-INX R1/R2 (SPRING Lab, IIT Madras, CC-BY-4.0); Kathbath (AI4Bharat IndicSUPERB, CC0); IndicVoices (AI4Bharat, CC-BY-4.0). Models: hishab (Conformer Large); SayedShaun (Whisper Medium mirror of tugstugi's competition model; wav2vec2 mirror of qdv206's); ehzawad (FastConformer-CTC; Voxtral Mini bn-BD, private, fine-tuned from mistralai/Voxtral-Mini-3B-2507, Apache-2.0). Author of this benchmark: Emrul Zawad.

Voxtral Mini bn-BD against the public models

Paired difference on the same recordings, WER points, 95% reference-clustered bootstrap (10,000 draws, seed 20260903)

Set	best public model	WER points	95% CI	paired interval
FLEURS test	Conformer Large (hishab)	+0.35	[-0.36, +1.06]	inconclusive
Vaani test	ehzawad FastConformer-CTC	-2.59	[-3.24, -1.95]	lower
SPRING-INX R1	ehzawad FastConformer-CTC	-2.35	[-2.96, -1.76]	lower
SPRING-INX R2	ehzawad FastConformer-CTC	+1.59	[-1.53, +5.71]	inconclusive
Kathbath known	Conformer Large (hishab)	+1.44	[+0.77, +2.12]	higher
Kathbath unknown	Conformer Large (hishab)	+2.03	[+1.37, +2.71]	higher
IndicVoices conv.	ehzawad FastConformer-CTC	+0.79	[-0.04, +1.56]	inconclusive

Voxtral Mini bn-BD (4.7B parameters) is a private fine-tune of mistralai/Voxtral-Mini-3B-2507; its card documents no training data, so every cell is U. It ran through the benchmark's harness like the public models (manifests, SHA-256 checks, asr_core path, batch 1, scorer) on the same A100, decoded by vLLM 0.31 with CUDA graphs: one piece per request, greedy, at most 500 new tokens per piece. On FLEURS test it takes 438 ms per clip, 5x the highlighted model. Like Conformer Large and Whisper Medium it writes numbers as Bengali words, which the scorer counts as errors against references that use digits. On SPRING-INX R2, 3 recordings ran to the 500-token cap repeating a word; they contribute 663 of its 8,143 word errors and stay in its 40.01% (without them 36.82% on 997 recordings, a conditional diagnostic, not a corrected score). Intervals are pointwise, not adjusted for multiple comparisons; an inconclusive result is not evidence of equivalence.